

Service agreements — getting the existing ones in, and handling them after

27 September 2026, from the FMS build; updated 29 September with Dennis's answers on 0 kr and contract types. How Jensen's existing service agreements get into the system alongside the bikes, whatever form they are in today (a spreadsheet row, a signed scan, a filled-in Word file, a verbal promise), and how the system handles them from then on. It builds on the modelling brief BRIEF-SERVICE-AGREEMENTS-2026-09 (agreements are per bike, not per customer). Read that one for the data model. This one is about the **sources and the import**, and it corrects the brief on price and terms. **Built on 29 September:** the per-bike model (§8, step 1) and the document upload (§5). The register import and renewal invoicing are not built yet.

1 • What has been said and decided so far

The 15 September meeting with Dennis (02:33–02:45):

- About 99 % of customers want an agreement. It is settled in the offer or the sales order, and sent after the build, once the frame numbers exist.
- Contract types **K1 / K3 / K5 / K10** (years). GPS bikes are signed for at least three years.
- *"I have a lot of service agreements, and I have maybe a handful signed."* The rest are verbal, worth hundreds of thousands of kroner a year, and **live in the spreadsheet**, which "has to come into the system" because the yearly invoicing runs from it.
- Renewals are invoiced **12 months ahead, a month before the anniversary**, and renew unless cancelled.
- Signing today: PDF → print → sign → scan → email. Dennis wants digital signing "for the future".
- The system should give each agreement a number. Dennis said he had sent the written agreement text.

What the system has today. One agreement per *customer*. Coverage comes from who owns the bike, and the fee is billed monthly in arrears (DECISIONS 2026-06). It was generated early and never discussed with Dennis. Production holds one test agreement, so changing it costs nothing yet.

The 27 September brief (not yet decided). An agreement is the document, and each covered bike is a line on it with its own start date and price. Coverage means the bike has an active line. Renewal invoices are drafted per customer × department EAN × month. The register's twelve month sheets (867 rows) become the lines. **Paper was never designed for:** the only step was asking Dennis for the text and for scans of the signed agreements (question C10).

What already exists to build on.

- The fleet import's **review step** (`scripts/import_fleet.py review`), which already parses the renewal schedule.
- **Provenance on imported rows** (`import_batch_id + import_row` , migration 102).
- A generic **attachments** table, not yet used for agreements.
- The **inbound pipeline's rule**: a model extracts, code matches deterministically, a person reviews, and nothing is written unseen.

2 · What we found in Dennis's files

In Documents/1-Projects/Jensen/ , received 7 June and not read until now:

- **Kommuneservice – final_v10_10år.dotx** , the agreement for municipalities.
- **Virksomheder – final_v7 1.docx** , the agreement for companies.
- **Extension of service agreement/...pdf** . The yearly "extension" is not a contract. It is an **e-conomic invoice** (no. 7114, Herlev Hjemmepleje):
 - two frame numbers in the text;
 - product JP-SERVFL "Forlængelse af serviceaftale(r)", 2 × 1 704 kr, for 13-05-26 to 13-05-27;
 - the department's EAN, and 30 days to pay.

What the templates say, and what it changes:

TOPIC	MUNICIPAL TEMPLATE	COMPANY TEMPLATE	CONSEQUENCE
Term	10 years	3 years fixed, extendable to 5 from purchase, then yearly at Jensen's discretion	K10 / K3 / K5 / K1 are real, and they differ by customer kind
Price	Per bike per month: 142 kr in year 1, 152 kr in years 2–5, 162 kr in years 6–10 , paid yearly in advance	<i>xxx kr</i> for the first 3 years, then <i>xxx kr</i> per year	The price steps with the bike's age ; a single frozen price per line (the brief) is not enough
Bike leaves (stolen, retired)	Moves to a replacement bike; if not replaced, the remaining years are invoiced up front	Moves to a replacement; if not replaced, the rest of the period is settled	Answers most of questions C6 and C7 to Dennis
Cancelling	In writing, 1 month's notice before a payment date	Same	Cancelling stops the <i>next</i> renewal, not the current year
GPS	Minimum 3 years, then continues until cancelled	Same	A line with GPS has its own binding end
What is covered	Punctures, tyres, tubes, chains, sprockets, bottom bracket, pedals, stand and other wear parts; e-bike motor / display / controller for 2 years; pick-up and return; labour	Same; battery warranty 2 years, not extendable unless agreed	This is the "covered or invoiced" rule for Finn's repairs
Not covered	Vandalism, self-inflicted damage (battery misuse, cables after a fall), missing own maintenance (air, chain oil)	Same	
Drive charge	<i>xxx kr</i> per visit inside the municipality, <i>xx kr</i> outside	Not in the template	A per-customer figure the system needs
Bikes listed	A table: make, type, frame number	Same	The frame number is the key between a document and a bike

But the spreadsheet does not follow the stepped price. Of the 867 renewal rows, 338 charge 1 704 kr (the year-1 price) and only 3 charge 1 824 kr (the year-2-to-5 price), although most bikes are well past their first year. So either the steps are not applied in practice, or older agreements used an older template. **Dennis answered (28 Sep): 1 704 kr is the real price every year, and there is no older template.** So the lines carry ONE frozen yearly price, as the brief had it; the template's steps are not modelled.

3 · The principle: three sources, one job each

SOURCE	IT DECIDES	IT DOES NOT DECIDE	HOW IT ENTERS
The register's month sheets	<i>What we bill</i> : which bike, its anniversary, this year's price, the EAN	Contract type, terms, who signed	A generated data migration, like the bikes (§4)
The documents (signed scans, filled-in Word or PDF files)	<i>The terms</i> : contract type, term, price steps, drive charge, signed date and signatories, contact person	Which bikes we bill this year	Attached to the agreement; a proposal is extracted and a person confirms it (§5)
e-conomic's invoice history (JP-SERVFL lines)	<i>Proof</i> of what was actually renewed, and at what price	Anything, on its own	Read-only cross-check (§5)

When they disagree: the spreadsheet wins for what is billed, the document wins for the terms, and every disagreement is **listed for Dennis, never resolved silently**. That is the same rule as the fleet import: unmatched rows become review lists, not guesses.

Consequence: **billing can go live from the spreadsheet without waiting for a single scan**. If only a handful of agreements are signed, the documents are evidence, not data. If Dennis has more than he remembers, step 5 grows without touching the others.

4 · Importing the agreements from the spreadsheet

Order: **bikes first**, then the agreement model, then this import.

1. **Group** the schedule rows into agreements: one per customer × department EAN (a municipality with five departments gets five agreements, because each is invoiced to its own EAN). Rows with a blank EAN (141 of 867) go under the customer, and are listed for Dennis.
2. **Match each row to a bike by frame number**. 656 rows match a bike in the fleet sheets; 211 do not. An unmatched row becomes no line: it goes on a list. It may be a bike the fleet sheets are missing, and inventing the bike would put a guess into the fleet.
3. **Each matched row becomes a line:**
 - start date = the purchase or delivery date, which sets the anniversary;
 - this year's price, from the row;
 - GPS when the price or the flag says so;
 - the years already invoiced (the X marks) kept as history.
4. **Every imported agreement is active and verbal** until a document is attached, and **carries no contract type**. Dennis (29 Sep): K1 / K3 / K5 / K10 are for agreements *from now on*; the existing ones were never typed, so none is recorded and none is guessed from the price or the customer kind. They renew one year at a time (to be confirmed, §7).
5. **A 0-kr row is an agreement that has ended** (Dennis, 29 Sep). It becomes an **ended** line, reason *ended*, end date *unknown*: never invoiced, and **the bike is not covered**, so Finn's repairs on it are invoiced. It is kept rather than dropped, because "we had an agreement until some year" is what a

customer will say on the phone. The X marks do not date the end: the sheets only track years 2–4, so a bike marked X through 2018 may have been billed after it. Listed for Dennis instead of guessed:

- **28 bikes at 0 kr whose customer sheet says *aftale: ja*** — the schedule wins for billing (§3), but a *ja* there may mean the bike should still be covered;
- **11 frames with both a 0-kr row and a priced row** — most likely a bike that was replaced or moved, the old line ended and a new one started;
- **13 rows with no price at all** — neither 0 nor a price, so neither invoiced nor ended until he says.

6. **Provenance:** agreements and lines carry `import_batch_id` and `import_row`, the same columns as imported bikes. “Imported” is a column rather than a note, and a bad batch can be found and undone.

7. The load is a numbered migration generated from the reviewed files, applied to production and the local copy and then queried. The same route as the bikes.

5 • Documents: upload, read, confirm (built 29 Sep)

Not blocking billing; done agreement by agreement, as documents turn up.

1. **Upload.** On the customer’s page, *Upload agreement*. On the phone, *Take photo* opens the camera, one photo per page and as many pages as the paper has; *Choose file* takes a PDF or photos. The files go to a private store and are never discarded: the signed paper is the legal record. A Word file is refused with “save it as PDF”. Uploading from an agreement’s own page attaches the paper to that agreement.
2. **Read.** The system reads the pages into a **proposal**:
 - the department and the contract type;
 - the signed date and the signatories;
 - the price, with ×12 if it is per month;
 - GPS;
 - **every frame number listed**;
 - anything to check (handwriting, unreadable parts).

It uses the extraction setting the inbox already has, so there is nowhere else to configure it. 3. **Match.** Code, not the model, matches each frame number to a bike: *this customer’s bike, another customer’s bike, a bike with no customer, not in the system, or already on an agreement*. A near miss (a letter O read for a zero) is offered as *Did you mean ...*, never ticked. 4. **Confirm.** Dennis chooses which of the customer’s agreements the paper belongs to, or *new agreement*, corrects the fields, and ticks the bikes. Only the customer’s own bikes that no other agreement holds start ticked. A bike on another agreement moves only when ticked. **Nothing is written before Confirm**, the same rule as calls in the inbox. 5. **e-economic cross-check.** Read the past JP–SERVFL invoices: – they name the frames and the period, so they show which bikes were really renewed, and at what price; – run it once before the first renewal run, as a before-and-after check. – **Blocker:** the system still holds only e-conomic’s *trial* grant, so the production grant has to be in place first.

Dennis’s step-by-step guide: GUIDE–DENNIS–AGREEMENTS–DA–2026–09 (PDF, Danish, with phone screenshots).

Dennis said there are “maybe 3 or 4” signed papers, so reading them was first left out. The owner then asked for it “smart”, and because the model reads PDFs and photos directly it cost little, so it was built.

6 • Handling after import

- **New agreements** are born from the offer or the sales order (*wants service agreement*), and carry a **contract type: K1 / K3 / K5 / K10 = the number of years the customer commits to**. The commitment is what the leave rule settles against (§2: remaining years invoiced up front); a GPS bike commits to at least 3. After the committed years, the line renews yearly until cancelled. They are pre-filled with what is known, sent after the build once frame numbers exist, and the document is generated from Dennis's text rather than a Word template. Signing stays print-and-scan until digital signing is chosen; the signed scan is attached as in §5.
- **Renewal** (planned in detail in `plan-renewal-invoicing`):
 - on the 1st of each month the system drafts the renewals for the next month's anniversaries, matching Dennis's month sheets: one draft per customer × department EAN, one line per bike, the period on the line;
 - the price is the line's frozen yearly price (no steps);
 - 30 days to pay, as the real invoices say, **not** the app's net 14;
 - lines with no price or no EAN are listed for Dennis, never guessed;
 - Dennis reviews and issues. **How it then reaches a municipality by EAN is open**: the FMS sends issued invoices to e-conomic as journal vouchers, which e-conomic cannot send electronically (see §7).
- **A bike leaves** (stolen, retired, cancelled): the line ends with a reason and date. The contract's rule applies: move it to a replacement bike, or settle the remaining term (municipal: remaining years up front). Nothing is credited automatically.
- **Cancelling** takes effect at the next payment date after a month's notice. The current year stays paid.
- **Repairs**: coverage is "the bike has an active line". What is covered is the contract's list (§2); everything else, and the drive charge, is invoiced.

7 • To find out

Answered by Dennis, 28 September:

1. Documents: **maybe 3 or 4**; the rest are verbal.
2. They **list frame numbers**.
3. **1 704 kr is the real price every year** — the template's steps are not charged.
4. **There is no older template**.
5. e-conomic: Dennis has a login for the API part. **Still needed**: someone with admin rights approves our app's install link, which produces the production grant token (it goes into the settings, never into chat).

Answered by Dennis, 29 September:

1. **A yearly price of 0 kr means the agreement has ended**; the bike is no longer invoiced (and so no longer covered). See §4, step 5.
2. **Contract types are for the future**: K1 / K3 / K5 / K10 say how many years the customer has committed. Existing agreements have no type.

Settled by the documents themselves (no need to ask): payment is 30 days (invoice 7114); invoices go per department EAN (the same invoice); what is and is not covered, cancelling, and a bike leaving (both templates, §2).

Still open — to Dennis:

- **Existing, untyped agreements:** they renew one year at a time and can be cancelled at any anniversary with a month's notice — so a bike leaving one settles nothing? (In effect K1.)
- **Where the commitment counts from:** when a K10 municipality adds a bike in year 4, is *that bike* bound for 10 years from its own delivery, or only until the agreement's 10 years are up?
- **Does the K-type change the price** of a new agreement? 1 704 kr (2 184 kr with GPS) for all four, or cheaper for a longer commitment? The register's other prices (2 284, 2 160, 1 200, 480 ...) are imported as they stand.
- **The three review lists** from §4 step 5 (28 *ja* at 0 kr, 11 frames twice, 13 without a price).
- **The drive charge** (*xxx kr* per visit in the template): the real figure, and is it per customer?
- **GPS without an agreement:** the register has a sheet for GPS subscriptions with no service agreement. Is that still invoiced, and should the system invoice it too?
- **The switch-over month:** from which month does the system draft the renewals instead of the spreadsheet? Both running for the same month is a double invoice to a municipality.
- Is the register or John's Trello export the source (question C1)?
- **Bikes with no EAN** (141 register rows): invoice them to the customer's own EAN?
- **The renewal invoice text:** copy invoice 7114 word for word?

Still open — to the owner (planning chat):

- **How a renewal reaches a public customer by EAN.** Today e-conomic sends it, because Dennis makes the invoice there. FMS invoices go to e-conomic as journal vouchers (DECISIONS 2026-07-09), and a voucher cannot be sent. The options are e-conomic draft invoices for renewals, OIOUBL sent by the FMS, or a list Dennis re-keys; they are compared in `plan-renewal-invoicing`, decision 1. The same gap applies to every FMS invoice to a municipality.

8 • Sequence and decisions

1. **Decide the model** (the brief), amended here with the contract type as committed years on new agreements (none on imported ones), ended lines for 0-kr rows, the bike-leaves settlement rule, and the GPS binding. One frozen yearly price per line, as the brief had it — no price steps. **Built 29 Sep** (migration 110; DECISIONS 2026-09-29).
2. **Import the bikes** (after Dennis's fleet answers).
3. **Import the agreements from the spreadsheet** (§4). Billing truth in place.
4. **Renewal invoicing**, checked against e-conomic's history on its first run. Planned: `plan-renewal-invoicing` (drafting can be built now; delivery by EAN waits on the owner's decision).
5. **Documents** (§5). Built 29 Sep, together with step 1.
6. **New agreements from the sales order**, then digital signing.

Rejected:

- *Documents as the source of the lines.* Most agreements have none, and billing would wait on paperwork that does not exist.

- *Reading writing directly.* A misread price is an invoice to a municipality.
- *The paper attached to the customer alone.* A customer has several agreements (one per EAN), so the paper is uploaded on the customer and confirmed onto one of them.
- *Guessing the contract type from the price.* The prices do not follow the template.
- *Waiting to import until the documents are gathered.* The money runs from the spreadsheet today.